THEOREM. *In an integral domain* A, *for every polynomial* P *in the variable* x *with coefficients in* A, $P \in \mathsf{A}[x]$, *and every* $r \in \mathsf{A}$, *there exist unique* $\mathcal{Q} \in \mathsf{A}[x]$ *and* $\mathcal{R} \in \mathsf{A}$ *such that*

$$P(x) = (x - r)\,\mathcal{Q}(x) + \mathcal{R},$$

where

$$\mathcal{R} = P(r).$$

○ 1. The existence

Let $P(x)$ be a polynomial in $\mathsf{A}[x]$ of degree n,

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^{n} a_i x^i,$$

where each coefficient $a_i \in \mathsf{A}$. Evaluating it at $x = r$ gives

$$P(r) = a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = \sum_{i=0}^{n} a_i r^i.$$

Consider the difference

$$P(x) - P(r) = \sum_{i=0}^{n} a_i x^i - \sum_{i=0}^{n} a_i r^i.$$

By the linearity of summation, the two expressions can be combined term-by-term. Moreover, the constant terms for $i = 0$ cancel out ($x^0 = r^0 = 1 \Rightarrow a_0 x^0 - a_0 r^0 = a_0 - a_0 = 0$). The difference $P(x) - P(r)$ becomes

$$\begin{aligned} \sum_{i=0}^{n} a_i x^i - \sum_{i=0}^{n} a_i r^i &= (a_0 - a_0) + \sum_{i=1}^{n} \Big(a_i x^i - a_i r^i \Big) \\ &= \sum_{i=1}^{n} a_i \big(x^i - r^i \big). \end{aligned}$$

Each term $x^i - r^i$ in the above sum is a difference of two i-th powers. The following lemma on the factorization of $x^n - c^n$ for any $n \geq 1$ shows that every such difference can be factored by $(x - r)$. Applying the lemma term-by-term exposes a common factor $(x - r)$ across all the terms, so it can be factored out of the entire expression.

Lemma. *For any integer $n \geq 1$, the difference of two n-th powers, $x^n - c^n$, can be factored by $(x-c)$. That is, there exists a polynomial $S_{n-1}(x,c)$ such that*

$$x^n - c^n = (x-c)\,S_{n-1}(x,c).$$

○ (by mathematical induction on n)

The **base case** $n=1$ holds since

$$x^1 - c^1 = (x-c)\,(1),$$

where the constant $1 = S_0(x,c)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $S_{k-1}(x,c)$ such that

$$x^k - c^k = (x-c)\,S_{k-1}(x,c).$$

Induction Step. To show that the lemma holds for $k+1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^k x$ and grouping as

$$x^{k+1} - xc^k + c^k x - c^{k+1} = \big(x^{k+1} - xc^k\big) + \big(c^k x - c^{k+1}\big).$$

Factoring out the common components from each group gives

$$x\big(x^k - c^k\big) + c^k(x-c).$$

Applying the inductive hypothesis to substitute $(x-c)\,S_{k-1}(x,c)$ for $\big(x^k - c^k\big)$ yields

$$x(x-c)\,S_{k-1}(x,c) + c^k(x-c).$$

Factoring out the common multiplier $(x-c)$ results in

$$(x-c)\Big(x\,S_{k-1}(x,c) + c^k\Big),$$

where $x\,S_{k-1}(x,c) + c^k = S_k(x,c)$ is a polynomial — since $S_{k-1}(x,c)$ is a polynomial by the inductive hypothesis, and polynomial rings are closed under multiplication and addition.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Explicitly, the $S(x,c)$ polynomials are

$$\begin{aligned}
1 \quad & = S_0 \\
x(1) + c^1 = x + c \quad & = S_1 \\
x(x+c) + c^2 = x^2 + xc + c^2 \quad & = S_2 \\
x\big(x^2 + xc + c^2\big) + c^3 = x^3 + x^2c + xc^2 + c^3 \quad & = S_3 \\
x\big(x^3 + x^2c + xc^2 + c^3\big) + c^4 = x^4 + x^3c + x^2c^2 + xc^3 + c^4 \quad & = S_4 \\
\cdots \quad & \\
x\big(x^{n-2} + x^{n-3}c + \cdots + xc^{n-3} + c^{n-2}\big) + c^{n-1} = x^{n-1} + x^{n-2}c + \cdots + xc^{n-2} + c^{n-1} \quad & = S_{n-1} \\
x\big(x^{n-1} + x^{n-2}c + \cdots + xc^{n-2} + c^{n-1}\big) + c^{n} = x^{n} + x^{n-1}c + \cdots + xc^{n-1} + c^{n} \quad & = S_{n}
\end{aligned}$$

In more compact notation with the summation symbol

$$S_n(x,c) = x^n + x^{n-1}c + \cdots + xc^{n-1} + c^n = \sum_{i=0}^{n} x^{n-i}c^i$$

and

$$x\,S_{n-1}(x,c) + c^n = x\sum_{i=0}^{n-1} x^{n-1-i}c^i + c^n = \sum_{i=0}^{n-1} x^{n-i}c^i + c^n = \sum_{i=0}^{n} x^{n-i}c^i = S_n(x,c)\,.$$

The exact factorization is $\qquad x^n - c^n = (x-c)\sum_{i=0}^{n-1} x^{n-1-i}c^i.$

Several interesting properties characterize $S_n(x,c) = \sum_{i=0}^{n} x^{n-i}c^i$ as a *complete homogeneous symmetric polynomial* in two variables :

- ✔ it is *symmetric*, since $S_n(x,c) = \sum_{i=0}^{n} x^{n-i}c^i = \sum_{i=0}^{n} c^{n-i}x^i = S_n(c,x)$,
- ✔ it is *homogeneous* of degree n, because the degree of each monomial $x^{n-i}c^i$ is exactly $n-i+i = n$ for all i,
- ✔ it is *complete* in the sense that it contains every possible n-th degree monomial in the variables x and c exactly once, each with a coefficient of 1.

The symmetry of $S_n(x,c)$ is already hinted at in the induction step. The proof inserts the zero $0 = -xc^k + c^kx$, but one may equally insert the zero $0 = -x^kc + cx^k$. Then $x^{k+1} - x^kc + cx^k - c^{k+1} = x^k(x-c) + c\big(x^k - c^k\big)$ and the resulting recurrence will be

$$S_k(x,c) = x^k + c\,S_{k-1}(x,c)\,.$$

The recurrences $x^k + c\,S_{k-1}(x,c)$ and $x\,S_{k-1}(x,c) + c^k$ are dual under the swap $x \leftrightarrow c$.

Returning to $\qquad P(x) - P(r) = \sum_{i=1}^{n} a_i\big(x^i - r^i\big),$

the lemma yields*

$$x^i - r^i = (x-r)\,S_{i-1}(x,r) \qquad \text{for every } i \geq 1\,.$$

* From a deeper algebraic viewpoint, $S_{i-1}(x,c)$ from the lemma belongs to the polynomial ring $\mathsf{A}[x,c]$. Replacing the indeterminate c by the element $r \in \mathsf{A}$ is the evaluation homomorphism $\mathsf{A}[x,c] \to \mathsf{A}[x]$. Thus $S_{i-1}(x,r)$ is a polynomial in $\mathsf{A}[x]$.

Substituting these factorizations term-by-term gives

$$P(x) - P(r) = \sum_{i=1}^{n} a_i \big(x - r\big) S_{i-1}(x,r) = \big(x - r\big) \sum_{i=1}^{n} a_i \, S_{i-1}(x,r).$$

Since each $S_{i-1}(x,r)$ is a polynomial in $\mathsf{A}[x]$ and polynomial sums are polynomials, the sum $\sum_{i=1}^{n} a_i \, S_{i-1}(x,r)$ belongs to the same polynomial ring $\mathsf{A}[x]$. Define

$$\sum_{i=1}^{n} a_i \, S_{i-1}(x,r) = \mathcal{Q}(x) \in \mathsf{A}[x] \, .$$

Hence $$P(x) - P(r) = \big(x - r\big) \mathcal{Q}(x)$$

or equivalently $$P(x) = \big(x - r\big) \mathcal{Q}(x) + P(r)$$

— the latter exhibits the intended representation with $\mathcal{R} = P(r) \in \mathsf{A}$. Ergo, such $\mathcal{Q} \in \mathsf{A}[x]$ and $\mathcal{R} \in \mathsf{A}$ exist. ●

It remains to show that $\mathcal{Q}(x)$ and $\mathcal{R}$ are unique.

○ 2. The uniqueness

Assume that there exist polynomials $Q', Q'' \in \mathsf{A}[x]$ and constants $R', R'' \in \mathsf{A}$ such that

$$P(x) = (x - r)\, Q'(x) + R' \quad \text{and} \quad P(x) = (x - r)\, Q''(x) + R''.$$

Subtracting the second representation from the first gives

$$(x - r)\, Q'(x) + R' - (x - r)\, Q''(x) - R'' = 0\,,$$

which simplifies to

$$(x - r)\big(Q'(x) - Q''(x)\big) = R'' - R'.$$

Evaluating this polynomial identity at $x = r$, the factor $(x - r)$ becomes $r - r = 0$, so the left-hand side vanishes and reveals* that the constant difference is zero :

$$0 = R'' - R',$$

that is $$R' = R''.$$

* Since $R', R'' \in \mathsf{A}$, the difference $R'' - R'$ is an element of A itself — equivalently, a constant polynomial in $\mathsf{A}[x]$. Accordingly, once evaluation at $x = r$ yields $0 = R'' - R'$, it follows that $R' = R''$ as elements of A. In other words, $R' = R''$ is an equality in A itself, not merely equality obtained after evaluating the polynomial identity at $x = r$.

Returning to the polynomial identity before evaluation and putting $R' = R''$ into it yields

$$(x-r)\big(Q'(x) - Q''(x)\big) = 0.$$

Because A is an integral domain, the polynomial ring $\mathsf{A}[x]$ is an integral domain as well and therefore has no zero divisors. Moreover, $x-r$ is a nonzero polynomial in $\mathsf{A}[x]$. Since its product with $Q'(x) - Q''(x)$ is the zero polynomial, the absence of zero divisors in $\mathsf{A}[x]$ implies that

$$Q'(x) - Q''(x) = 0,$$

giving

$$Q'(x) = Q''(x).$$

Thus

$$Q' = Q'' \quad \text{and} \quad R' = R'',$$

showing that the intended representation

$$P(x) = (x-r)\,\mathcal{Q}(x) + \mathcal{R}$$

is unique in the choice of $\mathcal{Q}$ and $\mathcal{R}$. ●

Definition. *Let $P \in \mathsf{A}[x]$ and $r \in \mathsf{A}$. If $P(r) = 0$, then r is called a* ***root*** *(or* ***zero****) of the polynomial P. Equivalently, r is a solution of the polynomial equation $P(x) = 0$.*

Corollary. *If $r \in \mathsf{A}$ is a root (solution) of the polynomial equation $P(x) = 0$, then there exists unique $\mathcal{Q} \in \mathsf{A}[x]$ such that*

$$P(x) = (x-r)\mathcal{Q}(x).$$

○ By the previous theorem, there exists unique $\mathcal{Q} \in \mathsf{A}[x]$ such that

$$P(x) = (x-r)\mathcal{Q}(x) + P(r).$$

Since r is a root, $P(r) = 0$. Therefore

$$P(x) = (x-r)\mathcal{Q}(x).$$ ●

In this case, $(x-r)$ is called a *factor* of $P(x)$, just as one says that a number b is a factor of another number a when b divides a. Hence every root $r \in \mathsf{A}$ gives a linear factor $(x-r)$ of the polynomial.

Remark. The above definition concerns roots that belong to the same coefficient domain A. However, a polynomial in $\mathsf{A}[x]$ may fail to have roots inside A and acquire them only in a larger domain or field containing A. For instance, $x^2 - 2 = 0$ has no root in $\mathbb{Q}$ but has the roots $\pm\sqrt{2}$ in $\mathbb{R}$. Whenever a root r exists in such an extension, the same theorem gives factorization $P(x) = (x-r)\mathcal{Q}(x)$ in the polynomial ring over that larger domain.